

INTRODUCTION TO DEMOGRAPHY

Demography is the scientific study of size, composition (structure) as well as change in the human population through the interaction of fertility, mortality and migration. Demography is dynamic and not static; it makes considerable use of mathematics and statistics. Demography involves the analysis of the determinants and consequences of the change in demographic phenomena i.e. fertility, mortality and migration.

DEMOGRAPHIC TERMS AND DEFINITIONS

Population size is the number in absolute terms of people in a defined territory or space. It can also be expressed in relative terms i.e. percentage.

Population Composition refers to the constituent elements that make up a population i.e. the make-up, structure or profile of the population e.g. age and sex. Others are marital status, race, ethnicity, occupation etc.

Population Growth or Change is the increase or decrease of population in absolute or relative terms.

Fertility is the number of births occurring in a population.

Mortality is the number of deaths occurring in a population.

Migration is the number of movements of people in and out of a particular population (area). It can be internal or international. Migration is also concerned with spatial distribution of the population in space.

SPATIAL DISTRIBUTION

- It is the location of the population in a space.
- It is the concentration of the population in a certain area.
- It is the dispersion of the population in a space.

ANNUAL POPULATION GROWTH RATE

The population can grow **geometrically** i.e. growing annually.

$$P_{t+10} = P_t (1 + 10)^t$$

It can also grow exponentially.

$$P_{t+10} = P_{t+10} e^{rt}$$

The process of population change can be expressed in the balancing of the population equation or balancing equation.

e.g.:

$$P_{2000} = P_{1990} + B_{(1990-2000)} - D_{(1990-2000)} + I_{(1990-2000)} - E_{1990-2000}$$

N.B: Immigration and Emigration i.e. **I** and **E** are used in case of **International migration**. In case of **Internal migration** use **In** and **Out** of a population area i.e. **I** and **O**.

DISTINCTION BETWEEN FORMAL DEMOGRAPHY AND POPULATION STUDIES

1) FORMAL DEMOGRAPHY

Formal demography is the aspect of demography which is concerned with the collection of statistical and mathematical analysis of demographic data. **Formal Demographers** are statisticians or mathematicians who deal with demographic variables in a mathematical way.

Examples of Formal Demography:

- Census taking.
- Examining the relationship between changes in life expectancy and age structure (using construction of life tables).
- Examining changes in birth rates as a function of change in the number of women in child-bearing years i.e. women between the age of 15 and 49 years, mortality and age structure or life expectancy.

2) POPULATION STUDIES

Population Studies is the inter-discipline approach to demography involving the biological and social sciences. It involves studies of the relationship between demographic and non-demographic variables.

- a) It involves the impact of non-demographic variables on demographic variables.

Examples:

- The impact or influence of income on fertility or mortality.
- The impact or influence of education on fertility and mortality.
- b) It also involves the impact of demographic variables on non-demographic variables.

Examples:

- Fertility and its impact on consumption patterns.
- Aging of a population and its impact on voting patterns.

REASONS FOR STUDYING POPULATION

- It helps us determine the number and distribution of people in a given area (spatial distribution).
- It helps us analyze the factors leading to population change (population dynamics).
- It determines the relationship between population change and socio-economic change.
- It helps in predicting of population changes and their consequences.

Demographic knowledge does assist both the private sector and government to place policies that are meant to benefit the welfare of the population. Virtually all policies related to education provision, employment creation, health care, food production among others depend on demographic knowledge.

The demographic Variable most important in making policies related to education is the **age structure**.

IMPORTANT SIDE NOTES:

↑ *Life expectancy* → *Population gets older and vice versa*

↑ *Fertility* → *Population gets younger*

- Relationship between non-demographic and demographic variables:

↑ *Income (economic variable)* → *Fertility* ↓ (*demographic variable*)

↑ *Income (economic variable)* → *Mortality* ↓ (*demographic variable*)

- *High education* → *delays child – bearing* → *fewer births* (E.g. comparison between a graduate and a primary dropout).

- *Knowledge on nutrition and health tips → reduces mortality*
- *Knowledge on contraceptives → reduces fertility*

- Relationship between demographic variables and non-demographic variables:

Fertility (demographic variable) → affects change in consumption patterns
(Economic variable)

Fertility (demographic variable) → age structure (demographic variable)
→ *Voting patterns (political variable)*

Voting patterns are influenced by age.

Migration → affected by income

E.g. out migration of working staff due to low income.

Fertility → affected by culture

- For others children are a source of insurance in old age.
- For others children are a source of labor.

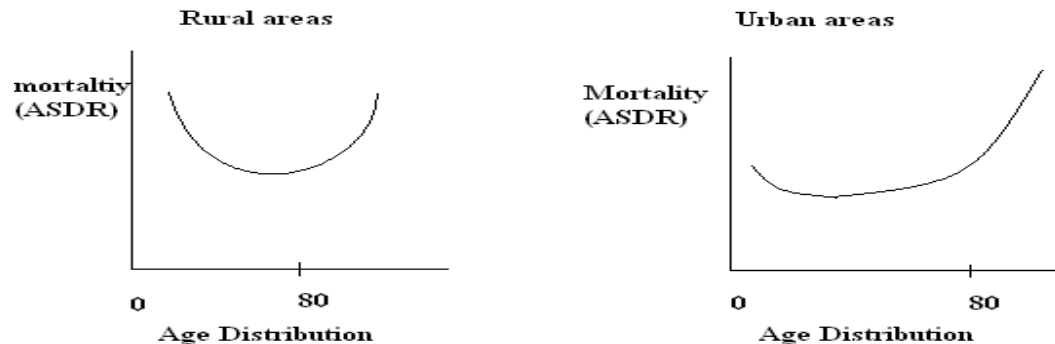
THE HISTORY AND DEVELOPMENT OF DEMOGRAPHY

➤ JOHN

GRAUNT

- John Graunt is generally referred to as the father of demography.
- He pioneered or initiated the numerical analysis of births and deaths.
- He obtained data (or figures) for what was known as **Bills of Mortality**. The Bills of Mortality were records which contained information derived from burials and christening (baptism). The records were compiled by the parish clerks. They began to compile these records during the period of the **Great Plagues of Europe** which occurred in the 16th and 17th Centuries. One of the plagues was referred to as the **Black Plague of Europe**.
- On the basis of this data, John Graunt attempted to discover uniformities or regularities or common patterns in vital statistics. **Vital Statistics** refer to vital events such as marriages, births and deaths.

- He further attempted to formulate some laws of population growth. He did this in a book he published in **1662** called **Natural and Political Observations Made upon the Bills of Mortality**.
- His findings were confined to England and London in particular. He discovered that in London, deaths exceeded births. This meant that there was virtually no growth in the population. On the other hand, in rural areas, births exceeded deaths. As a result there was high growth of population in the rural areas which exerted pressure on the available resources thereby stimulating rural-urban migration.
- He also found out that at birth, the number of male infants exceeded the number of female infants.
NOTE: Empirical evidence has proved this even today. This discovery was both a biological and demographic discovery. However, with the passage of time, sex ratio changes, probably due to the fact that females tend to outlive males i.e. females have a higher life expectancy as compared to males. Perhaps this can be attributed to the occupation that males take-up and their manner of lifestyle (i.e. smoking and drinking) which exposes them to a high risk of death.
- John Graunt also established the distributive age pattern to mortality. He established a **U-Shaped Curve**.



ASDR: Age Specific Death Rate

- When conditions improved, the curve became **J-Shaped**. Mortality continues to rise in old age even when conditions improve due to the biological deterioration of people.
- He also did the classification of deaths by cause and by geographical variation as well as by seasonality. By classification it meant that by major causes of deaths and the distribution of these deaths.

- He also pioneered the construction of a **Crude Life Table** which was a precursor to the modern life table.
- John Graunt was the first person to estimate the population of London. He also identified fertility, mortality and migration as the major dynamics of in population change.
- Graunt pioneered the use of statistical methods in evaluating demographic data.
- **Edmond Halley** improved on John Graunt's life table.

➤ **THOMAS MALTHUS (1766-1834)**

- Thomas Malthus was both a clergyman and an economist.
- In **1798**, he wrote a book called **An Essay in Population**. The writing of this book was as a result of the French Revolution that took place in France and culminated into the over-throw of the monarchy. This took place during the reign of **Louis** the 16th and **Marie Antoinette** his wife. Most of the high class were scared and were gotten rid of because the poor considered them parasites. This scared most of the high class Europeans including those in Britain like Thomas Malthus. Thus the British started discussing ways of improving the conditions of the poor as a way of preventing a similar revolution.
- In his book, Thomas Malthus was concerned about cutting the population of the poor because the poor were such a threat to the upper class of whom Thomas Malthus was part of.
- In his book he presented some basic arguments. He observed that food production tended to increase by arithmetic progression (i.e. 1, 2, 3, 4 ...). In other words, food production was increasing very slowly. Whereas, the population was increasing in geometric progression (i.e. 1, 2, 4, 8). In other words, the population was increasing very fast.
- According to Malthus, the rapid increase in population was attributable to the unrestrained reproductive habits of the poor who bred like animals to him. This rapid increase of population of the poor jeopardized the food supply. He therefore, suggested the need to slow down the population growth of the poor segments of the population.
- He made suggestions on how to tackle this problem.

- He came up with what he called **moral solutions** in which he suggested that the poor should restrain from their sexual life by delaying marriages and abstaining from sex.
- Furthermore, he identified the checks on population growth. These are **Preventive Checks** and **Positive Checks** on population growth.

PREVENTIVE CHECKS

- By Preventive Checks, Thomas Malthus meant the voluntary mechanisms (means) of controlling births. Among the means he suggested that one of the ways of controlling births was through prolonged celibacy and chastity. Celibacy being refraining from marriage and Chastity being abstaining from all sexual intercourse.
- He excluded things like adultery, prostitution, sexual deviation (homosexuality and lesbianism), birth control (use of contraception) and abortions because, as a clergyman, these were not moral or proper ways of preventing births.

POSITIVE CHECKS

- Positive checks can be categorized into two categories which are **Man-made (wars)** and **Natural Disasters (famines, epidemics e.t.c)**.
- By positive checks, Thomas Malthus meant involuntary means of controlling births. Among these, he included wars, epidemics, famines and food shortages. So that if people didn't control births voluntarily through preventive checks, positive checks were evitable.
- He referred to food shortages as the ultimate positive check.
- He discouraged improving the social conditions of the poor because to him, improving the social conditions of the poor could encourage them to give more births.
- He argued that high demand for food was a good incentive to increasing economic activities and food production in particular. He therefore, suggested that the rich should consume more food so that more food can be grown for the economy.

➤ KARL MARX (1818-1883) AND FREDERIECH ENGELS (1820-1895)

- Marx and Engels were political and economic philosophers.
- These two philosophers did not view population as a constraint to economic growth in a way that people like Malthus did. For that reason they rejected the Malthusian Theories.
- Both believed that each economic system has its own laws of population.
- A distinction was made by Marx and Engels between two broad economic systems. Namely the **Capitalist System** and the **Socialist System**.

- According to the two theoreticians, in the capitalist system, they believed that population can be manipulated in such a way that population could be changed to a surplus population or an industrial reserve army.
- Marx and Engels when talking about the industrial reserve army or surplus population said, “There were three forms which it could take”.

THREE FORMS OF SURPLUS POPULATION OR INDUSTRIAL RESERVE ARMY

▪ FLOATING POPULATION

- This according to Marx and Engels, were that part of the surplus population replaced (displaced) by machinery during the process of industrialization resulting from technological innovations.

▪ LATENT POPULATION

- According to Marx and Engels, this was the agricultural population which was being displaced by mechanization and commercialization of the agricultural sector. The displaced people became potential migrants who then became susceptible to the whims of the capitalists.

▪ STAGNANT POPULATION

- This consisted of those people with highly irregular employment and low living standards e.g. casual workers because these are easy to replace when fired.

THE CAPITALIST SYSTEM

- Ownership of the means of production is in the private sector.
- Prices of goods and services are determined by the laws of supply and demand.
- Marx was saying that the capitalists would create artificial employment.

THE SOCIALIST SYSTEM

- Ownership of the means of production is in the hands of the state.
- Prices of goods and services are determined by the state.

- According to Marx, the surplus population could not be manipulated but it could be fully utilized through the use of labor as a means of production.
- Marx and Engels also believed that the increasing number in population was not a threat to food supply. It was seen as a population of potential producers.

THE GROWTH AND DEVELOPMENT OF DEMOGRAPHY IN THE 19TH CENTURY

Apart from the contributions of Marx and Engels, there was growing interest in Demography and increasing concern in population issues and problems in the 19th Century. In **1982**, Demography received explicit international recognition at the **4th International Conference in Hygiene,**

Geneva,

Switzerland.

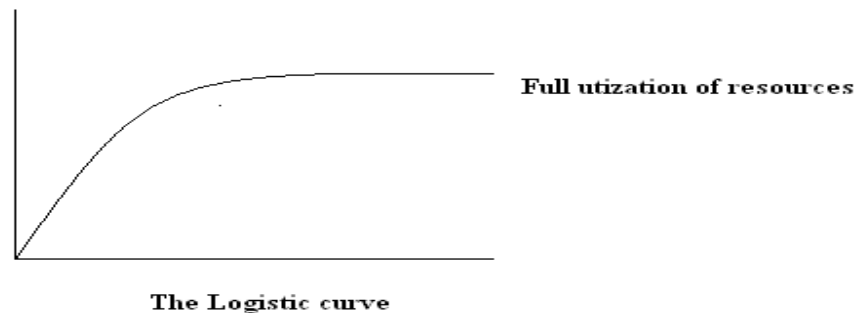
Further, impetus to Demography was the reception that was accorded to it as a discipline in **1885**. That is it was given formal recognition by the **National Statistical Institute.**

DEVELOPMENT OF DEMOGRAPHY IN THE 20TH CENTURY

Some of the impetus to the development of Demography as a scientific discipline included the ground-breaking work by two American scholars namely **Raymond Pearl** and **Lowell Reeds.**

These two refined the application of the logistic curve to understand issues of population growth

In effect, the work by Reeds and Pearl was inspired by the studies of animal population. According to Reeds and Pearl, population reaches an upper limit as it approaches the full utilization of its resources. This conclusion emerged from the observation that species of animals grow rapidly when placed in a limited environment with ideal food supply and space.



The Logistic Curve is a mathematical curve. The population reaches on 90-Upper limit and upon reaching this upper limit, the population ceases to grow. The Logistic Curve is criticized for not taking into account:

- The fact that technology changes can increase food supply.
- The diffusion of contraceptives and family planning practices.
- The restrictive regulation policies that some countries may impose on migration to prevent influx.

The Logistic Curve was improved upon by demographers like **Alfred Lotka** and **Yule** so that it could fit contemporary issues.

INSTITUTIONALIZATION OF DEMOGRAPHY

Much of the pioneering work which contributed into the acceptance of Demography as a discipline was done in the early 20th Century mostly in the United States of America.

During the first quarter of the 20th Century, the most competent students of population studies were **Dublin, Lotka, Raymond Pearl** and **Lowell Reeds**.

These were concerned primarily with issues of health, actuarial studies and biological sciences.

The scope and orientation of population studies was affected significantly by the entrance of social scientists with a background of sociology and economics. These include scholars like **P.K. Whelpton** and **Frank Notestein**.

The diversity in the background led to the interdisciplinary approach in the study of population

studies.

Since the early 20s, Demography grew rapidly as a field of interrelated sciences ranging from economics, biology, actuarial studies, sociology etc. By **1927**, the **First World Conference** was held in **Geneva, Switzerland**. Since then the co-operation between demographers and international organizations started to grow. The first mode of development was the sponsorship by the **League of Nations** of a series of population studies under direction of **Frank Notestein**. **Frank Notestein** was the **1st Director** at office of **Population Research, Princeton University**. This office was established in **1936**. In subsequent years, central research and population studies training programs were established in many universities in the USA and later spread to Europe. In the early **1950s**, a **Population Council** was established by **John Rockefeller**. The population council is devoted to the study and research of population studies. The **United Nations** through the **United Nations Population Fund (UNFPA)**, Formerly known as the **United Nations population Fund of Population Activities (UNPPA)**, which also contributed significantly to the establishment of **Training and Research Institute** in **1986**. **The UNZA Demography Department** was established in and has produced over 200 graduates in Demography. The UNFPA has also established Training and Research Centers in **Africa (Ghana)**, **Asia (Bombay)**, **Latin America** (**Chile**).

RELATIONSHIP BETWEEN DEMOGRAPHY AND OTHER DISCIPLINES

Demography differs from natural sciences such as Physics, Chemistry and Biology in terms of **subject matter** but in **similar** in terms of **methodology**.

Demography involves the **study of people** whereas, natural sciences involves the study of **objects**.

However, both demography and natural sciences deal with observable Phenomena. Births, deaths, movements, electricity, sound, rusting, reproduction, respiration are all observable phenomena.

Thus demography is similar in methodology to natural sciences in that they both deal with empirical phenomena (i.e. observable and experienceable phenomena). Both involve descriptions, explanations and predictions. Demography as a scientific study is a highly inter-discipline social science. It is synthetic in

nature i.e. ii theories on elements and methods of different disciplines such as economics, sociology, mathematics, biology, medicine etc.

▪ **RELATIONSHIP WITH ECONOMICS**

Economics is a social science concerned with the description and analysis of the production, distribution and consumption of goods and services. Economics is related to demography in many ways.

Economic conditions in a given country can influence demographic events.
Example:

Economic theories and models are sometimes used to describe demographic phenomena.

- ✓ Example, the cost-benefit model analysis can be used to describe reproduction or fertility as regards to decision making at household level. Just like may have to weigh the costs and benefits of buying a commodity, a couple may have to weigh the costs and benefits of having an extra child.

COSTS

- Maintenance costs.
- Health costs.
- Educational costs.
- Opportunity costs (e.g. education will have to be sacrificed by the wife).

BENEFITS

- There may be psychological benefits such as happiness.
- Old age insurance i.e. the children will look after the parents when they are old.

✓ **The Law of Supply and Demand:**

- This can be applied in the marriage market.
- For instance, more eligible young men versus few eligible young women. This can reduce the probability of marriages thereby reducing fertility
- E.g. more women versus few men will lead to high fertility.

▪ DEVELOPMENT PLANNING

Population is a very critical variable in planning.

✓ PLANNING FOR EDUCATION

In planning for education, age of entry into the school system is very cardinal. To determine this, we use the skills and expertise of a demographer who does the population distribution by single years.

AGE	POPULATION
1	
2	
3	
5	
7	

For example, the age of entry into school in Zambia is seven years. Using the number of 7years old, the government or private can determine the number of schools, classes, desks, books etc that are needed per province, district, town etc. They can calculate the cost of how much each 7 year old will incur if enrolled into the school system and multiply by the total number of 7 year olds to come up with the total cost. As these primary school children progress, their number will be affected by mortality, migration, drop-out rates and failure rates i.e. the transition from primary school is affected by demographic variables.

Demographers can also project population and thus is necessary for future planning.

✓ PLANNING FOR HOUSING

In housing development, we also look at the age distribution. People begin to own houses in a given range of years mostly between 20-30 years of age. This is

determined by the age at marriage because most people enter the marriage market between that age ranges.

✓ **PLANNING FOR HEALTH CARE**

Knowledge on population composition is very important in planning for health care. One needs knowledge on the population of people who are:

- Under-five years so as to know how many under-five clinics are needed.
- between 15-49.
- Over 65 years of age.

✓ **PLANNING FOR EMPLOYMENT**

- We look at the population broken down by age group.
- We look at the age of entry into the labor market. (This 15 years according to the UN).
- It is this number that will determine the demand for employment. Thus the government will know how many jobs ought to be created.

▪ **SOCIOLOGY AND DEMOGRAPHY**

Sociology is the scientific study of the society, social institutions that the society as well as the social relationships, among other things such as culture.

Socio-cultural conditions can influence mortality, fertility and migration.

- Culture (beliefs, norms, values, and customs) » Mortality
For example, Polygamy increases the probability of one contracting HIV/AIDS. Circumcision if not hygienically done may lead to the contraction of diseases such as HIV/AIDS.

Sexual cleansing also increases the probability of one contracting HIV/AIDS.

- It is these diseases that raise mortality.
- Female genital mutilation has implication on maternal mortality.
- However, reduces chances of contracting HIV/AIDS and therefore, lowers mortality.

Chastity reduces the probability of HIV/AIDS.

Small family norms bring about child-spacing which reduces maternal mortality among women in child bearing years.

Early marriages increase fertility.

Celibacy and chastity lower fertility.

- People have a tendency to migrate to areas where they may have similar culture and therefore, it would be difficult for them to adapt for example, people would rather migrate to a place where they speak the same language than to a place where communication would be difficult due to the diversity in language.
- Some migrate because they run away from oppressive cultural practices where they are compelled to conform to their culture against their will thus they run to those societies which are more liberal and tolerant.
- Demography (population change) can also influence socio-cultural change in variables.
- Fertility/mortality » age structure » change in lifestyle, tastes, norms, values, attitudes.
- When fertility increases, holding mortality constant, there is the younging of population. This change in age structure brings about change in socio variables such as lifestyle, tastes, attitude etc.
A young population may be characterized by the presence of cinemas, disco houses, sports stadiums as well as many entertainment structures.
- When fertility decreases, holding mortality constant, there is the aging of the population.
- Migration has also brought about inter-tribal marriages.

▪ **PSYCHOLOGY AND DEMOGRAPHY**

Human behavior can influence attitude towards certain demographic phenomena.

Example: Attitude towards the use of contraceptives.

▪ **MEDICINE AND DEMOGRAPHY**

Medicine is linked to demography in the area of reproductive physiology.

- Example the impact of breastfeeding on birth intervals. Breastfeeding extends the birth interval thereby reducing the number of births a woman can have i.e. breastfeeding has an involuntary effect or contraceptive effect on fertility.
- Medicine has contributed immensely to the development of contraceptive technology. This has given demographers insight into the natural methods in the regulation of fertility.

- The areas of **morbidity**, **epidemiology** and **biostatistics**.
Morbidity is the study of the causes of death.
Epidemiology is concerned with the distribution of diseases.
Biostatistics is the statistical study of causes of deaths, incidences of deaths as well as distribution of deaths.

▪ REGIONAL STUDIES/PLANNING

This field of study links demography to geography. It is concerned with the geographical distribution of the population. It deals with city and regional planning. For example, the determination of zoning regulations depends on the distribution of the population.

Zoning is done in terms of industrial, residential and commercial areas. For example, some areas may be zoned strictly for industrial purposes or as residential areas or strictly for commercial purposes.

In Zambia, zoning regulations that were proposed the Doxiades Reports have not been followed due to a political interference. For instance, Kanyama areas which was supposed to be an industrial area but is now a residential area, Chibolya was supposed to be an industrial/commercial area.

▪ ECOLOGY AND ENVIRONMENTAL SCIENCE

Ecology is the study of the interaction of human population and the environment. As the human population increases, it encroaches forests and other animal habitats. There should be a **balance of nature** or **ecological balance** to ensure harmony between forests, animals and human beings. Industrialization disturbs this ecological balance through pollution. Good examples in Zambia include Mufulira, Kitwe, Kabwe and Kafue. Kitwe and Mufulira are adversely affected by sulphur dioxide.

▪ DEMOGRAPHY AND MATHEMATICS

In Demography there is a very considerable use of mathematical models and methods. E.g. life tables and statistic population models depend almost entirely on mathematics. Similarly, population projections depend almost entirely on mathematics.

▪ **DEMOGRAPHY AND LAW**

Sometimes laws, legislations and legal institutions may influence population. For example, the one-child policy in China has a very big impact on the fertility. Others include laws on abortions, (immigration) or migration laws. For example, in Zambia, in the early 1940s, there were certain laws (Chitupa) that were passed to prevent migration from rural to urban areas. In South Africa, they had influx control laws (Butustom) policies which were internal migration laws.

Laws of citizenship also influence international migrations.

▪ **GENDER STUDIES AND POPULATION**

Role differentiation of men and women in terms of prescribed roles and achieved roles. Gender studies also look at the empowerment of women. The relative power (or position) between men and women in society has an impact on fertility. For example, if women take a subordinate position in society, this may raise fertility. Conversely, if a woman is empowered, the woman takes part in decision making in the home, fertility is likely to be low because she takes a greater part in reproductive decisions.

▪ **POLITICAL SCIENCE AND DEMOGRAPHY**

Distribution of power is supposed to be done in relation of population size for example, the creation of parliamentary constituencies. The determination of voting age depends on demography. The allocation of resources and representations also depends on demography.

CONTEMPORARY PROBLEMS AND ISSUES

An African-American social scientist **Lester Brown** working for the **World Watch Institute** has observed that the population is a phenomenon. He says population growth can compound, magnify and even create a variety of social, economic, political and many other problems.

1. FOOD SECURITY

If population growth exceeds food production, there will be food security.

This is referred to as the **Malthusian Trap**. In reality, food production must exceed population growth.

2. POLLUTION

Increase in population both in terms of population size and population growth means that the population has to be catered for in terms of employment creation. Thus, industries such as mining industries, manufacturing industries are set up to provide employment. These industries have a potential of polluting the environment. It may also lead to other problems of deforestation and environmental (deforestation may result from increased demand for human settlement areas which is brought about by pollution). An example of a country with this scenario is Mexico.

3. INFLATION

Inflation is a situation where demand exceeds supply i.e. “too much money chasing too few goods”.

Scarcity induced inflation takes place when population increases fairly rapidly without corresponding increase in production of goods and services.

4. INCOME

Rapid population growth can also affect economic growth. If population growth increases faster than economic growth, there will be a decrease in the amount of goods and services available to the population. This may lead to low per capita income holding all other things equal.

5. UNEMPLOYMENT

An increase in population tends to accelerate the size of the labor force. If there is no subsequent increase in job opportunities or job creations, the result is an increase in the levels of unemployment. The opposite is true i.e. if a lot of jobs are being created than the number of people getting into the labor force due to low fertility, unemployment levels decrease e.g. the situation in Germany.

6. SOCIAL PROBLEMS (CRIME, PROSTITUTION ETC)

A rapid increase in population will cause unemployment levels to increase which will result into crime, prostitution as a survival strategy. Crime rates usually go up where there is an increase in unemployment.

7. LITERACY

When the population is increasing faster than the government's capacity to build more schools, fewer people will be able to access education thereby increasing the level of illiteracy. **Literacy** is the ability to read and write at least a paragraph in one's own language and **illiteracy** is the opposite of literacy. Because of high fertility, you have high volume of children thereby.....

8. INDIVIDUAL FREEDOM

When there is increase in population, you end up with more people requiring space and resources. Where this happens, the government has to come up with rules to ensure and regulate the use of the limited space and resources for the common good of all. This eventually can cause encroachment on one's freedom if this problem of space resource allocation is not sorted out.

9. INFRASTRUCTURE PROBLEMS

The consequences of high population growth may be manifested in form of water supply problems and transportation problems. This is because some infrastructures were initially meant for a smaller population and with increase in population these problems increase proving a threat to the few resources available.

10. HOUSING AND PROPERTY

Increase in population induces increase in the demand for housing. When this happens, the cost of land, real estate and the cost of construction of houses also go up. Because the majorities have limited financial capacity to build decent houses due to the increase in costs, the construction of substandard houses eventually increase thereby increasing shanty compounds.

11. ENERGY

Population increase also impacts on energy in the sense that every additional number of the population requires energy to prepare food and for warmth. It is energy that propels the economy.

Increase in population results in increase in demand for energy which results in the shrinking in energy reserves.

Examples of energy include oil, electricity, nuclear energy, coal and solar energy. Julim Simon, an American economist has written a book in which he invalidates the Malthusian theories. He argues that the ultimate resource is human ingenuity. That regardless of the increase in population, human beings are still good to find ways out of the problems.

THE DEMOGRAPHER IN THE WORK PLACE

Demographics is applied demography. It involves application of demographic information and methods in business administration etc.

Application of demography is relevant in:

- Human resource management
- Marketing
- Investment
- Management demographics
- Insurance
- Sociology
- Education
- Health
- Public services
- Political planning
- Campaign strategizing

❖ HUMAN RESOURCE PLANNING

In discussing this, we distinguish between internal and external demographics. Human resource management involves looking at the wellbeing of workers. **Internal Demographics** is concerned with the demographic make-up or profile of the workers. The human resource management is concerned with qualities and characteristics such as age, gender, marital status. This is because knowledge of demographic make-up of employees will influence decision making in terms of how best to fulfill their needs. For example, knowledge of age will make it possible to formulate policies of retirement. The age and education profile will give you the need of training for employees. Knowledge of age, gender and marital status helps formulate policies related to maternity leave. The determination of the number of people to retire makes use of demography using life tables. **External demographics** involve the demographic trends outside a company. For example, high fertility and low mortality impacts a company in terms of labor force.

High fertility + Low Mortality = High labor force

The size of the labor force determines the conditions of service of workers.

Low fertility > Mortality > Low labor force > High retirement age

❖ INVESTMENT DEMOGRAPHICS

Investment use demographics to pin-point or locate areas of potential or actual economic growth. They are aware that population is an important factor in social change and therefore, opportunities.

High fertility > Younging population > High demand for entertainment goods and services.

Making sound investment decisions involve for-casting into the future and assessing the type of investment to be made. Harold Levine, a vice-president of E.F Hutton in the USA says that the 20% of decision making in investment is demographic, 30% is economic and 50% is on company's capability. To him, demographic planning is a cornerstone for corporate finance.

❖ DOMESTIC DEMOGRAPHICS

Dynamic demographics focus on demographic trends on the local or domestic market that may influence production of goods and services. Knowledge of demographic trends may guide businesses in making appropriate site selections, identifying market areas, fore-casting demand sales and estimating market share.

○ **DEMOGRAPHIC PATTERNS AND INVESTMENT PATTERNS**

Changes in the population structure, changes in patterns of migration and changes in the level of urbanization may also influence investment patterns. These changes may result in changes such as increasing in purchasing power, increase in the size of the middle class and increase in expatriate migration. These changes may influence the establishment of certain industries and structures to cater for these changing patterns.

○ **LOCAL INVESTMENT OPTIONS**

Demographic changes especially changes in the population structure may also influence the investment options e.g. investment in stocks and bonds (shares), treasury bills. Stocks and bonds are long-term investments whereas treasury bills are short-term investments. The demographic implication is that it is more meaningful for younger people to invest in long-term investment like stocks and bonds whereas older people would rather invest in treasury bills which have short-term benefits.

○ **INTERNATIONAL DEMOGRAPHICS**

Investors also respond to global demographic change. Many companies look around the world for investment opportunities. To do so, they use global demographic trends.

❖ **MARKETING DEMOGRAPHICS**

Marketing is one area which uses a lot of demographic information and methods. In marketing, we make a distinction between **segmentation** and **targeting**.

▪ **SEGMENTATION**

This refers to the manufacturing and packaging of goods and services or the provision of products that appeal to identifiable socio-demographic groups. In other words, making the

production of goods and service that will appeal to a specific group e.g. setting up cinemas (ster-kinekor and block busters) is tailored to appeal to younger generations. Boutiques are packaged to appeal to a certain gender, sex, social class, age etc.

▪ **TARGETING**

Targeting involves picking out particular socio-demographic characteristics of people who might purchase what you might offer and appealing to their consumer tastes and behaviors as reflected in those characteristics.

By advertizing by appealing to the characteristics of the people, targeting uses statistics of age, gender, marital status, occupation and education. These demographic characteristics reflect certain tastes. Ethnicity/race is also another demographic variable that is targeted in marketing.

▪ **CLUSTER MARKETING**

This involves targeting neighborhoods which share socio-demographic characteristics. It also has the element of geographical location, consumer tastes and demands collerate with socio-demographic characteristics of certain neighborhoods.

❖ **INSURANCE**

Insurance companies thrive on the sale of such things as life insurance. Life insurance is sold on the basis of premium. The determination of the amount you pay depend demographic information namely age. The younger the person, the lower the premium paid. Insurance companies need to have an understanding of life tables that function like the life expectancy.

❖ **PLANNING FOR SOCIAL SERVICES**

Planning for social services is given the jurisdictions of local agencies like the municipal council. To make efficient decisions, these agencies make use of demographic information e.g. water distribution, road construction and sewerage.

❖ EDUCATION

Demand for the provision of education services are determined by the demographic profile of a nation. School enrollment depends on fertility and other demographic trends. Demographers can project the size of education needed in the country.

❖ HEALTH

How many maternity and under-five clinic facilities to be built depend on the number of women in reproductive years. Clinics for old people can be planned using demographic information. Gender, sex and age influence decisions on health services.

❖ POLITICAL PLANNING

The creation of constituencies in constituency planning largely depends on the population distribution. The population structure determines the distribution of power. For example in Zambia:

LUSAKA	KITWE	NDOLA
Kanyama	Kwacha	Chifubu
Matero	Chimwemwe	Kalulushi
Kabwata	Wusakile	Ndola Central
Munali	Nkana	Bwana Mukubwa
Lusaka Central	Kanfinsa	
Chawama		
Mandevu		

Towns with a larger population tend to have more constituencies. The other use other use is in campaign strategy. The voting age population within a particular

constituency is a very key thing in winning an election. Demographic variables like age, tribe, ethnicity and gender can influence voting. Geo-demographics is the tendency of politicians to appeal to voters in one district with whom they share socio-demographic characteristics. The candidates are packaged in such a way that appeals to a certain categories of people. It is the determination of the demographic composition of people in a geographical area. You also have to look at certain demographic variables like age and gender. Males are more likely to vote than females and the young are more likely to vote than the aged. Opinion polls are used to determine voter preferences based on demographic information. It is a research method of collecting information to find out what voter preferences are. The use of demographic information is important with legislative analysis e.g. the age of retirement is influenced by life expectancy in a country. Thus the demographic changes can influence the enactment laws.

CAREERS IN DEMOGRAPHY

- **Research**
- **Lecturing and teaching**

GOVERNMENT

- **Health - Biostatistics**
- **CSO - Census**
- **Ministries (education, labor) –Planning, agriculture, finance.**

CITY MUNICIPAL COUNCIL

- **Planning for housing, water distribution, sewerage, road construction etc.**

INTERNATIONAL ORGANISATIONS

- **UNICEF – Population division of the UN (New York)**
- **UNFPA**

PRIVATE ORGANISATIONS

- **World Vision**

BUSINESS INDUSTRY

- **Insurance – Marketing – Human Resource Management**

INTRODUCTION TO COMMON CONCEPTS AND BASIC MEASUREMENTS IN DEMOGRAPHY

THE BASIC TOOLS

- Count
- Rates
- Ratios
- Proportions (percentages)

COUNT

This refers to absolute numbers of a population or any demographic event occurring in a specific period of time.

RATES

Rates refer to the frequency of demographic events in a population in a specified period of time. A distinction is made in demography between crude rates and specified rates. **Crude rates** are rates computed for the entire population. **Specific rates** are rates that are computed for specific sub-groups of the population.

RATIOS

This relates or expresses the relation of one sub-group to another sub-group in the same population.

PROPORTION

This is a relation of a population sub-group to the entire population. A proportion can be expressed as a decimal or percentage. A **percentage** an expression of a proportion in terms of a hundred units.

THE POPULATION STRUCTURE

SIZE OF A POPULATION

Example, the population size of Zambia in 2007 was 12.5 million.

STRUCTURE OR COMPOSITION OF A POPULATION

1) SEX STRUCTURE (COMPOSITION)

To understand this we use the sex ratio. The **sex ratio** is the number of males per hundred females.

$$\text{Sex Ratio} = \frac{\text{no. of males}}{\text{no. of females}} \times 100$$

e.g.:

$$\frac{4,946,298}{4,936,293} \times 100 = 100.1$$

In 1996, the sex ratio was 96 meaning there were 96 more females than males. Percentages/proportions are important in analyzing the population distribution.

PERCENTAGE DISTRIBUTION OF THE POPULATION

This is done by disaggregating the population by age group, marital status, and occupation.

0 – 4	-	-
5 – 9	-	-
10 – 14	-	-
80+	-	-
Total	100	100

The population distribution can be expressed graphically or in terms of population pyramids.

2) THE DEPENDENCY RATIO

Simply refers to the ratio of young people under 15 years and old people above 65 years to the economically active (middle-aged) people between 15 – 64 years.

$$\text{Dependency Ratio} = \frac{\text{under 15 yrs} + \text{old (65 yrs+)}}{\text{economically active (15 – 64 yrs)}}$$

3) THE MEDIAN AGE

This is the age which divides the population into two equal parts. This means 50% are below this age while the other 50% of the population is above that age. The median age is used to measure the aging of populations.

Young	Under 20 yrs
Intermediate	20 – 29 yrs
Old	30 and above

(High fertility)

AGE CLASSIFICATION

Child hood infancy	0 – 9 years
Adolescence	10 – 19 years
Early adolescence	10 – 14 years
Late adolescence	15 – 19 years
Youth	15 – 24 years
Young people	10 – 24 years
Middle age	40 – 65 years
Old age	Over 65 years

Classifications according to the United Nations

CONCEPTS AND MEASUREMENTS OF POPULATION CHANGE

○ MORTALITY

Fertility refers to the actual reproductive performance of a population. A **live birth** occurs when a fetus in whatever gestation age exits the maternal body and subsequently shows any signal of life, such as voluntary movement, heartbeat or pulsation of the umbilical for however brief a period of time (according to the UN). **Fecundity** refers to the physiological capability of couples to reproduce. **Fertility** is also the number of live births in a population; fertility is affected by fecundity. Other factors affecting fertility are:

- **Fecundity**
- **Age of marriage**
- **Availability and use of family planning (contraception)**
- **Economic development**
- **Status of women**
- **Age – structure (this refers to the balance of males and females in different age groups).**

MEASURES OF FERTILITY

The most common measure of fertility is the crude birth rate.

1. CRUDE BIRTH RATE

Crude birth rate simply indicates the number of live births per 1000 of the population.

$$\text{Crude Birth Rate (CBR)} = \frac{\text{No. of births}}{\text{Total population}} \times 1000$$

e.g.:

$$\text{India (1983) CBR} = \frac{25,745,000}{735,596,000} \times 1000 = 35$$

In 1980, the CBR for Zambia was 50. The crude birth rate is crude because it includes babies or people in the menopause in the denominator.

Constants have to be decreased when dealing with events that are common (more frequent) and they have to be increased when dealing with events that are rare.

2. THE GENERAL FERTILITY RATE (GFR)

The **GFR** is the number of live births per 1000 women aged 15 – 49 years in a given area at a given time.

$$GFR = \frac{\text{No. of births}}{\text{Women (15 – 49)}} \times 1000$$

The GFR is an improvement over the CBR.

E.g.:

$$\text{Puerto Rico (1980) GFR} = \frac{73,060}{826,690} \times 1000 = 88.4$$

Zambia (1990) GFR = 185 and in 2000, 152.

3. THE AGE SPECIFIC TERTILITY RATE (ASFR)

This is a more refined than the General Fertility Rate. These are rates obtained for specific age groups and are very useful in making comparisons of fertility behavior in different age groups.

$$AFSR = \frac{\text{No. of births at age 'x'}}{\text{No. of women at age 'x'}} \times 1000$$

$$\text{E.g.: ASFR} = \frac{\text{No. of births (20–24)}}{\text{No. of women (20–24)}} \times 1000$$

$$\text{Japan (1983) ASFR} = \frac{274,911}{3,882,924} \times 1000 = 70.8$$

Malawi (1977) ASFR was 280 in the same age group. Zambia (2000) in the same age group had 277.

Enrollment in local colleges, 2005

AGE	LIVE BIRTHS	WOMEN	ASFR (ZAMBIA)
-----	-------------	-------	---------------

15-19	-	-	.1407
20-24	-	-	.2768
25-29	-	-	.2692
30-34	-	-	.2317
35-39	-	-	.1748
40-44	-	-	.0833
45-49	-	-	.0301

4. THE TOTAL FERTILITY RATE (TFR)

The TFR is based on the ASFR. It is the average number of children that will be born to a woman during her life time if the woman experiences the ASFRs.

$$TFR = \text{Sum of ASFR} \times 5 \text{ (the size of the interval age group)}$$

In 2000, TFR was 6 for Zambia.

5. THE GROSS REPRODUCTIVE RATE (GFR)

The GFR is the average number of daughters that a woman will bear by the end of her reproductive age considering prevailing ASFR in a given year. It literally measures reproduction i.e. a woman reproducing by giving birth to a daughter.

6. NET REPRODUCTION RATE (NRR)

The NRR is the average number of daughters that will be born to a woman if she experiences the prevailing ASFR and mortality in a given year. The NRR tends to be than the GRR.
Zambia (2000)

TFR	=	6
GRR	=	2.5

NRR = 1.7

7. COMPLETED FERTILITY (PARITY) MEAN NUMBER OF CHILDREN EVER BORN

The **Mean number of children ever born** refers to women in a particular age group. **Parity** refers to the number of live births a woman has ever had in her child bearing age.

$$\text{Parity} = \frac{\text{No. of Children ever born (CEB) to a woman}}{\text{No. women in that age group}}$$

Enrollment in local colleges, 2005

AGE GROUP	WOMEN	CHILDREN	PARITY
<i>Undergraduate</i>			
15-19	-	-	0.3
20-24	-	-	1.4
25-29	-	-	2.7
30-34	-	-	4.1
35-39	-	-	5.4
40-44	-	-	6.4
45-49	-	-	6.8

In Zambia, a child ever born (CEB) parity for Zambia was as above as.....
Parity refers to life time fertility.
 Parity increases with age whereas ASFR fast increases then decrease.
 Parity is also referred to as **Completed Fertility Rate** or **Completed Family Size**.

CFR and **CFS** is the mean number of children per women past the limit of child-bearing i.e. past menopause. To get this you use parity for 45 – 49.
 CFR and CFS tends to be close to the total fertility rate.

○ CONCEPTS AND MEASURES ASSOCIATED WITH MORTALITY

Mortality refers to deaths as a component of population change. The rate at which death occurs is linked to many factors such as sex, age, race/ethnicity, occupation and social class. The incidence of death in a population reveals much about the population's standard of living, standards of health care and hygiene.

MEASURES OF MORTALITY

i. CRUDE DEATH RATE

The CDR is the number of deaths per 1000 population in a given year.

$$CDR = \frac{\text{No. of deaths}}{\text{Total population}} \times 1000$$

For example:

$$\text{Israel (1983) CDR} = \frac{27,818 \text{ deaths}}{4,037,620} \times 1000 = 6.9 = 7$$

In other words, there were 7 deaths for every one thousand Israelis in 1983. Zambia (1990), CDR was 18. This means there were 18 deaths for every 1000 Zambians. In 2000, the CDR was 20.

ii. AGE SPECIFIC DEATH RATE (ASDR)

ASDR is a more refined measure of mortality because rates are obtained for specific age groups. In some cases this can also be done by sex.

AGE	DEATHS	POPULATION	ASDR
0-4	-	-	-
5-9	-	-	-

$$ASDR = \frac{\text{Deaths in age group}}{\text{Population in age group}} \times 1000$$

E.g.:

$$\text{Bulgaria (1989) ASDR (20 - 24)} = \frac{541 \text{ deaths in (20 - 24)}}{625,626 \text{ people in (20 - 24)}} \times 1000 = 0.9$$

In other words, there was 1 death per every one thousand Bulgarians aged between 20 and 24 in 1981.

iii. INFANT MORTALITY RATE

The IMR is the number of deaths to infants under age one per 1000 live births in a given year.

$$\text{IMR} = \frac{\text{No. of deaths to infants under age 1}}{\text{Total live births}} \times 1000$$

E.g.:

$$\text{Sweden (1984) IMR} = \frac{594 \text{ infants dying}}{93,792 \text{ live births}} \times 1000 = 6.3$$

In other words, there were 6 deaths to infants under age 1 for every 1000 live births in Sweden in 1984.

For Zambia (1996), IMR was 109. This means there were 109 deaths to infants under age 1 in every 1000 live births in Zambia in 1996. These reflect the standards of living and standards of health and hygiene. In Zambia 2000, IMF was 110 deaths.

iv. CAUSE SPECIFIC DEATH RATE

CSDR are expressed in terms of deaths per 100,000 of a population.

$$CSDR = \frac{\text{No. of deaths to cause } (x)}{\text{Total population}} \times 100,000$$

$$\text{E.g.: Peru (1980) } CSDR = \frac{16,480}{17,779,500} \times 100,000 = 92.7$$

In other words, in Peru 93 people in 1980 died of pneumonia per 100,000 of a population

v. PROPORTION DYING OF A SPECIFIC CAUSE

It is an expression of deaths from a specific cause as a percentage of all deaths.

$$PDSC = \frac{\text{No. of deaths from cause } (x)}{\text{Total population}} \times 100$$

$$\text{E.g.: Peru (1980)} = \frac{16,480}{105,325} \times 100 = 16\%$$

CHILD MORTALITY RATE- measures the number of deaths to children under the age of five. In Zambia 2000, it was 82.

UNDER FIVE MORTALITY – It refers to the number of deaths among children below the age of 5 years. In Zambia 2000, it was 162.

vi. MATERNAL MORTALITY RATE (MMR)

MMR is the number of women who die as a result of children bearing in a given year per 100 thousand births in that year. Maternal deaths are deaths caused by complications of child bearing.

$$MMR = \frac{\text{No. of maternal deaths}}{\text{Total no. of live births}} \times 100,000$$

Maternal mortality is an average event hence we use a large constant.

$$\text{For USA (1983), } MMR = \frac{290}{3,614,000} \times 100,000 = 8$$

There were 8 maternal deaths for every 100 thousand live births in the USA in 1983. For Zambia (1996) had MMR = 649

vii. LIFE EXPECTANCY

Life expectancy is an estimate of the average number of additional years a person can expect to live given the prevailing age specific death rates. Life expectancy can be calculated after each year. The most commonly used life expectancy is the life expectancy at birth. e_0 Is the symbol for the life expectancy at birth.

The Greek alphabet

AGE	e_x
0	50 years
1	-
2	-
3	-

Life expectancy is computed on probabilities of dying and

viii. MORBIDITY

It refers to the incidence of disease and illness in a population

MEASURES OF MORBIDITY

- ✓ **THE INCIDENCE RATE** – Is the number of persons contracting a disease during a given period of time per 100,000 of a population.

$$IR = \frac{\text{During a given time period}}{\text{Total population at risk}} \times 100,000$$

$$\text{e.g.: Canada (1985)} = \frac{2,280}{25,858,500} \times 100,000 = 9$$

9 people per 100,000 of the population contracted rubella.

- ✓ **THE PREVELANCE RATE**

It refers to the number of persons having a particular disease at a given point in time per 1000 of a population. It consists of all previously existing cases plus newly developing ones.

$$PR = \frac{\text{No. of persons with disease}}{\text{Total population at risk}} \times 1000$$

✓ CASE RATE

It refers to the number of reported cases of a specific disease per 100 thousand of a population during a given year.

$$ER = \frac{\text{No. of reported cases}}{\text{Total population at risk}} \times 100,000$$

$$\text{e.g.: Venezuela (1984) CR} = \frac{12,088}{18,600,000} \times 100,000 = 65$$

This means that there were 65 reported cases of malaria.

✓ THE CASE FATALITY RATE

It is a proportion of persons contracting a disease and dying from that disease.

$$CFR = \frac{\text{No. of persons contracting \& dying from the disease}}{\text{No. of persons developing the disease}} \times 1000$$

○ MIGRATION

Migration is the movement of the population across a specified boundary for the purpose of residing in that place; a movement from a point of origin to a point of destination.

Internal migration involves movement within a country; it is movements between different areas within a country. You can have micro-internal migration within a district.

International migration involves the movement between countries.

INTERNAL MIGRATION

There are various types of internal migration. These are:
>>**Rural – Urban** is where you move from a place defined as rural to a place defined as urban.

>>**Urban – Rural** is movement from an urban area to rural areas e.g. one retires going to the village.

>>**Rural – Rural** is movement from one rural area to another rural area.
>>**Urban – Urban** is the movement from one urban area to another.
The reason why people move depends on pull and push factors.
Pull factors are factors that pull people to a particular area. They mostly consist of economic factors.

Push factors are factors that compel people to move out of an area e.g. absence of economic opportunities.

We distinguish internal migration between **in-migrants** (people coming into a district) and **out-migrants** (people moving out).

INTERNATIONAL MIGRATION

International migration takes two major forms; **the brain drain** and **the refugee problem**.

○ THE BRAIN DRAIN

People move to another country in search for job opportunities.
Brain Drain is the movement of skilled people across countries in search of job opportunities.

○ THE REFUGEE PROBLEM

This is the movement of people across international boundaries to seek peace or refuge due to political instability in their own countries.

We distinguish international migration between **immigrants** (people who come into a country from another country) and **emigrants** (people who leave a country and go to another country).

Among immigrants, we differentiate between **expatriates** and **residents** and **citizens**.
Expatriates normally come into a country on short-term contracts.
Some expatriates can be transitioned to residents when they stay in a country for a long period of time because they can then apply for citizenship.

Expatriates → Residents → Citizens

MEASURES OF MIGRATION

a) IMMIGRATION RATE

This is the number of immigrants arriving at a destination for every 1000 people at that destination.

$$IR = \frac{\text{No. of immigrants}}{\text{Total population at destination}} \times 1000$$

b) IN-MIGRATION RATE

It is the number of in-migrants at a destination divided by 1000 people at that destination

$$\frac{\text{No. of in – migrants}}{\text{Total population @ that destination}} \times 1000$$

For Sweden 1984:

$$IR = \frac{31,486}{8,336,597} \times 1000 = 3.8$$

There were 4 people coming into Sweden for every 1000 Swedes.

c) EMIGRATION RATE

It is the number of emigrants departing an area of origin per 1000 population at that area of origin in a given year.

$$ER = \frac{\text{No. of emigrants}}{\text{Total population at area of origin}} \times 1000$$

$$\text{Sweden (1984) } ER = \frac{22,825}{8,336,597} \times 1000 = 2.7$$

In other words, 3 Swedes left Sweden for every 1000 Swedes in 1984.

d) OUT-MIGRATION

RATE

$$\frac{\text{No. of out – migrants}}{\text{Total population}} \times 1000$$

e) NET MIGRATION RATE

This is the net effect of immigration and emigration. It is also the net effect of in-migration and out-migration.

$$\text{Net migration rate (NMR)} = \frac{\text{No. of immigrants} - \text{No. of emigrants}}{\text{Total population}} \times 1000$$

$$\text{Internal - NMR} = \frac{\text{No. in - migrants} - \text{out - out migrants}}{\text{Total population}} \times 1000$$

$$\text{Sweden 1984, NMR} = \frac{31,486 - 22,825}{8,336,597} \times 1000 = 1$$

For every one thousand Swedes, one person came into Sweden. NMR can be expressed as an increase or decrease per one thousand of the population in a given year.

MEASURING AND UNDERSTANDING POPULATION STRUCTURE AND POPULATION CHANGE

Population has three components namely **births**, **deaths** and **migration**.

THE BALANCING EQUATION

The balancing equation is one way of depicting change in the population. It is the most basic method of calculating numerical change in the population over a period of time.

$$P_t = P_0 + (B - D) + (I - O)$$

P_t Is the population at the later date.

P_0 Is the population at an earlier date.

B Is birth.

D Is death.

I Is the in-migration and

O is out-migration.

$$\text{e.g.: } P_{2000} = P_{1990} + (B_{1990-2000} - D_{1990-2000}) + (I_{1990-2000} - O_{1990-2000})$$

The equation is used to project population change.

Australia (1983)

$$15382000_{83} = 1,118,000_{82} + 242,000_{82-83} - 113,000_{82-83} + 93,000_{82-83} - 18000_{82-83}$$

Developed countries have reliable and accurate information hence the balancing equation.

NATURAL

INCREASE

Natural increase is the surplus or deficit of births over deaths in a population over a given period of time.

$$NI = B - D$$

Where **NI** is natural increase, **B** is births and **D** is deaths.

RATE OF NATURAL INCREASE

The rate of natural increase is the rate at which the population is increasing or decreasing in a given year due to the surplus of births over deaths expressed as a percentage of the base population. It also does not include the impact of immigration or emigration.

$$RNI = \frac{\text{Births} - \text{Deaths}}{\text{Total mid-year population}} \times 100$$

Mid-year population is the average of the base year population and the later year population.

$$\text{For Australia 1983, } RNI = \frac{242,000 - 113,000}{15,280,000} \times 100 = 0.8\%$$

The rate of NI can also be calculated from the birth and death rates.

$$RNI = \frac{\text{Birth rate} - \text{Death rate}}{10}$$

For Australia 1983,

$$RNI = \frac{15.8 - 7.4}{10} = 0.8\%$$

This was growth due to the difference between births and deaths.

POPULATION

GROWTH

RATE

The growth rate is the rate at which a population is increasing or decreasing in a given year due to natural increase and net migration. It is also expressed as a percentage of the base population.

$$GR = \frac{242,000 - 113,000 + 75,000}{15,280,000} \times 100 = 1.3$$

The GR can also be calculated from knowledge of RNI and NM.

$$\begin{array}{rclcl} GR & = & RNI & + & NM \\ = & & 0.8 & + & 0.5 \\ = & > & 1.3 \end{array}$$

$P_t = P_0 e^{rt}$ To calculate the growth rate you make r the subject of the formula.

The GR is normally higher than RNI because it includes both RNI and NM. When GR is declining, it might not necessarily mean that the population is declining but that the population is growing at a slower rate. A negative GR means that an area is losing population.

DOUBLING TIME OF POPULATION

Growth which is expressed as a percentage is sometimes not helpful or informative. The best way to find out how the population is growing is to find out how long a population will take to double in size.

$$2P_0 = P_0 e^{rt}$$

A simpler way is to divide:

$$\frac{70}{\text{Growth rate (\%)}}$$

$$\text{For Australia, } \frac{70}{1.3} = 53.8 \text{ years}$$

$$\text{For Zambia, } \frac{70}{2.8} = 28 \text{ years}$$

It means that Zambia's population would double in 28 years.

POPULATION AND SOCIAL STRUCTURE

Social structure refers to stable patterns of interaction in a society. The stable patterns of interaction are defined by institutions in the society, statuses, roles and groups.

SOCIAL INSTITUTIONS

It is simply a set of stable rules and relationships that regulate social activities. The major institutions that regulate school activities are:

- Economy
- Family
- Education
- Religion
- Medicine

These social institutions work hand in hand with culture in order to shape the social life of society.

STATUS

Members of society who interact in the context of these institutions hold social positions in an institution.

Attached to each status, may be a number of roles that one is supposed to play. Roles are referred to as responsibilities or behaviors, privileges that are expected of individuals occupying certain social positions.

SOCIAL GROUPS

It refers to two or more people who interact with each other on a regular basis and also share common expectations and a common identity e.g. of social groups are tribes, social clubs i.e. UNZAPOPSA, UNZABECA, Lions Clubs etc. Roles and statuses and individual and the ways individuals interact are to a very great extent conditioned by demographic variables like age, sex and marital status.

AGE

It determines the status one has in society and even the social roles one can play in society. It plays a vital role in the lifestyle of individuals because it influences your entrance to:

- Education

- Labor force
- Voting

THE DEMOGRAPHIC TRANSITION THEORY

This is the most widely accepted theory of population growth. This theory states that both a population's fertility and mortality will decline from high to low levels as a result of economic influences. The decline in mortality precedes the decline in population growth during a period of transition.

STAGE 1

High birth rate + high death rate = no increase in population.

STAGE 2

High birth rate + falling death rate = high growth in population.

STAGE 3

Declining birth rate + relatively low death rate = slowed population growth.

STAGE 4

Low birth rate + relatively low death rate = very low population growth.

Some demographers think a 5th Stage; the argument is that: When fertility goes down, a slow rate of population growth can turn into a negative one.

POPULATION AND ECONOMIC DEVELOPMENT

Economic development is measured in terms of rise in per capita income or increase in production of goods and services. This is by the World Bank. The UNDP has come up with the HDI on the argument that wellbeing depend on income only. HDI looks at longevity, access to clean water, education etc.

This is complimentary between population and economic development. In turn economic factors influence demographic factors.

Demographic factor



Economic factors

INFLUENCE OF ECONOMIC DEVELOPMENT ON POPULATION

Economic development influence population through the pull and push factors.

Pull factors are factors that attract people to move towards certain destinations e.g. movement to areas of better economic development.

Push factors are factors that push people out of their areas of origin.

CONSEQUENCES AT ORIGIN

- Loss of skill and labor
- Reduced productivity
- Lowering of living standards

CONSEQUENCES AT DESTINATION

- Increase in population
- Unemployment if there is no corresponding job creation
- Increased in uncontrolled urban settlements (shanty compounds).

IMPACTS OF ECONOMIC DEVELOPMENT ON MORTALITY

Mortality decreases when a country develops due to rise in per capita income which leads to higher purchasing power.

- This means increased access to health facilities.

- Improvement in medical technology.
- Improvement in water supply and sanitation.
- Improvement in education and peoples' appreciation of high standards of hygiene increases.

INFLUENCE OF ECONOMIC DEVELOPMENT ON FERTILITY

Economic development is associated with lower fertility due to:

- Expansion of the education system, more women is enrolled in the schools, the mean age of marriage is raised and women are less likely to have more children.
- Expansion of education makes the small "family norm" to become a norm "standard".
- Knowledge of contraceptives due to
- Careers may be preferred by women other than child-bearing.

DEMOGRAPHIC FACTORS INFLUENCING ECONOMIC FACTORS

➤ GROWTH RATE

If population growth exceeds economic growth rate, the result is poverty. If on the other hand, economic growth exceeds population growth, people's standards of living will rise.

➤ POPULATION SIZE

If population size exceeds available resources, then the country cannot give its population the best , facilities.

When the population is too big in relation to available resources, it becomes difficult to attain what is called **an optimum population.**

Optimum population is the population that gives the greatest comfort and standards of living to the inhabitants of a country; if a population grows beyond the optimum, that population risks exhausting the available resources by exceeding its carrying capacity.

Carrying capacity is that size of the population that could theoretically be maintained indefinitely at a given level of living.

➤ AGE STRUCTURE

This depends on how fertility and mortality interact. E.g.:
High fertility + Low mortality → Young age structure

- Increase in dependence burden.
- Consumption exceeds savings thus retarding economic growth.
- More young people get into the labor force; unemployment rises if there is no corresponding increase in job creation.

Low fertility + Low mortality → Aging of population → More old people

- Low savings because you have more retirees who need to be catered for. These retirees consume more than they save.
- Tax rates may go up so as to cater for the retirees.

POPULATION AND SOCIETY

Society is made up of people and social institutions, statuses, roles and groups.

SOCIAL INSTITUTIONS

These are simply a set of stable rules and relationships that regulate social activities in which members of society engage to meet certain needs. The main social institutions are the economy, family, education system and religion; social institutions work together with culture to shape the social life of a society. **Culture** refers to rule and norms that regulate the way people behave. Members of society hold certain social positions in the society. Status is a social position you hold in a social institution. In holding that status, certain rules are expected of them that hold them. The roles, statuses of individuals and the way they interact are defined by demographic variables like age and sex.

AGE

Age normally determines the social status and roles of an individual. It also plays a critical role in the life cycle of individuals because it influences important landmarks in

one's life.
It determines entry into the education system, entry in marriage, labor force etc.

Low fertility + Low mortality → Old population

In most societies, social statuses are given according to age rankings. If there is low fertility and low mortality, this leads to an increase in the number of old people. This may shift the balance of power to older people. You might have the emergence of a system of government called **gerontocracy**. **Gerontocracy** is a system of government dominated by older people where greater power in decision making is given to older people; the ideas and values of older people dominate. This may slow down the process of social change.

High fertility + Low mortality → Young population

This may result into generational conflict between the old and the young people if the old people hold on to power longer. Younger generations are more receptive to technological change, new lifestyle, cultural values and consumer tastes. This adaptability to change may accelerate the process of social change. This acceleration of social change may translate into rapid economic growth which may counteract the effect of high population growth.

SEX

Status of women

High fertility + Low mortality

High fertility imprisons a woman to a depressed status characterized by issues of child caring and nursing. This is a situation so made worse especially in traditional circles where early marriages are encouraged. Early marriages reduce a women's chance of continuing with education thus it gives a woman a low social status.

Low fertility + Low mortality

Low mortality increases the chances of survival of children. If children survive at an increasing rate, it reduces the load of women to have more children to compensate for those that die. This enhances a woman's status, opens up opportunities of career advancement, takes up more social roles and responsibilities because she is not preoccupied by issues of child-bearing.

STATUS OF WOMEN AND ITS IMPACT ON MORTALITY AND FERTILITY

The status of women may impact on fertility and mortality in a different way. Improved social status of women may lead to lower fertility. When a woman has a high social status, she tends to evaluate the cost and benefits of having more children. She may evaluate the impact of fertility on her career thus she's more likely to look at an additional child as a cost. Increased participation in the labor force compels them to have fewer children. Similarly, when a woman has a prestigious job that gives them a high social-economic status, it means the job demands more of a woman's time resulting in lower fertility. Women with high socio-economic status have more say in matters concerning when to have a child, how many children the couple should have and this reduces fertility. This empowering of women is a very powerful contraceptive method. They will even space the few children that they have and this reduces infant mortality. It also reduces the risk of maternal mortality. High women that have high socio-economic status have higher income and hence they feed their babies with more nutritious foods thereby reducing infant mortality rate.

MIGRATION AND SOCIAL CHANGE

Social change means change in lifestyle, behavior, values, norms etc. When people move from one area to another, they have to change their lifestyle and values. If they move from rural to urban areas, other social institutions like school and the economy. The move transforms a villager into an urban person with new roles, values, behaviors and practices. Thus when a person moves into an area, they will have to adapt to the environment by conforming to the norms, values and practices of that area.